

Solution Requirements

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The current version of the system does not require user authentication or login. Users can directly access the application and enter soil and environmental parameters to receive machine learning-based crop recommendations.</i>	Medium
2	Authorization levels	<i>The system provides different access levels. Farmers can submit agricultural parameters and receive recommendations, while administrators can manage datasets, retrain the ML model, and maintain the application.</i>	Medium

3	External interfaces	<i>The system interacts with the Crop Recommendation Dataset, the trained Machine Learning model (.pkl), Flask web application, and optionally weather information services for future enhancements.</i>	High
4	Transactions processing	<i>The system accepts user input (N, P, K, temperature, humidity, pH, rainfall), validates the data, preprocesses it, generates crop recommendations using the trained ML model, and displays the results.</i>	High
5	Reporting	<i>The system displays crop prediction results and may maintain prediction history or logs for future analysis and model improvement.</i>	Medium
6	Business rules, etc.	<i>Input values must fall within valid agricultural ranges. The prediction is generated only after successful validation. The recommendation is based on the trained machine learning model using environmental and soil parameters.</i>	High
7	Compliance to laws or regulations	<i>The system stores only essential agricultural information, follows basic data privacy principles. Any collected user data should be handled securely and ethically.</i>	Medium

8	<i>Other</i>	<i>The system provides an easy-to-use web interface, supports responsive design, and allows future integration with fertilizer recommendations, weather forecasting, and market price information.</i>	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should quickly process user inputs and generate crop recommendations.</i>	<i>Prediction displayed within 2 seconds after submission.</i>
2	Scalability	<i>The application should support increasing numbers of users and larger agricultural datasets without significant performance degradation.</i>	<i>Supports 500+ simultaneous users with consistent response time.</i>
3	Security & Data Privacy	<i>User inputs and prediction records should be protected from unauthorized access. Input validation should prevent malicious data submission.</i>	<i>Secure input validation and restricted access to stored data.</i>

4	Reliability & Availability	<i>The system should consistently provide accurate recommendations and remain available during normal operation.</i>	99% system availability during deployment.
5	Usability & Accessibility	<i>The web application should provide a simple, responsive, and user-friendly interface suitable for farmers with minimal technical knowledge.</i>	<i>Responsive UI compatible with desktop and mobile browsers; intuitive navigation.</i>
6	Maintainability & Portability	<i>The system should use modular Python code, making future maintenance, updates, and deployment on different platforms straightforward.</i>	<i>Modular Flask application with reusable code and deployment support on Windows/Linux.</i>